

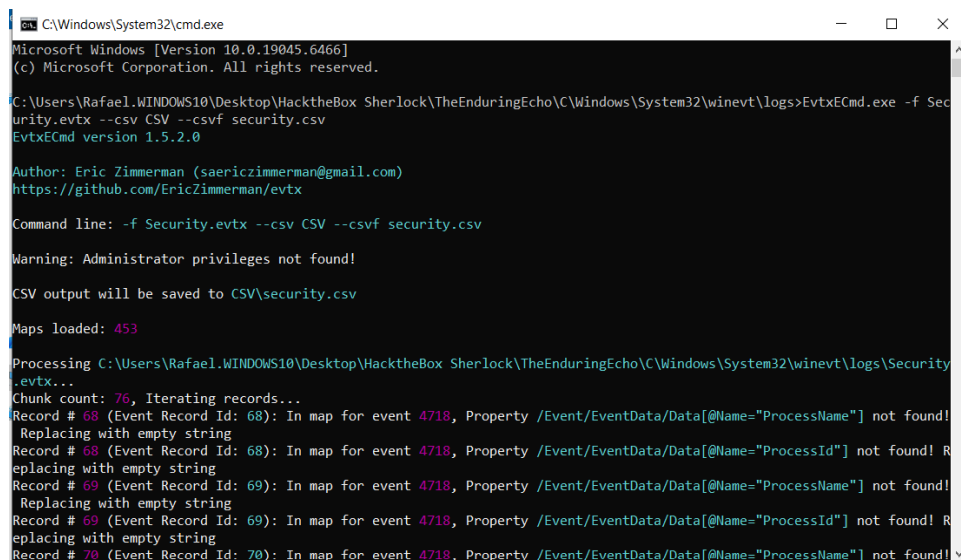
# HACK THE BOX SHERLOCK - DFIR

## HOLMES 2025 3: THE ENDURING ECHO

### Sherlock Scenario

LeStrade passes a disk image to Holmes. It's one of the identified breach points, now showing abnormal CPU activity and anomalies in process logs.

1. What was the first (non cd) command executed by the attacker on the host?



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Rafael.WINDOWS10\Desktop\HacktheBox Sherlock\TheEnduringEcho\C\Windows\System32\winevt\logs>EvtxECmd.exe -f Security.evtx --csv CSV --csvf security.csv
EvtxECmd version 1.5.2.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/evt-x

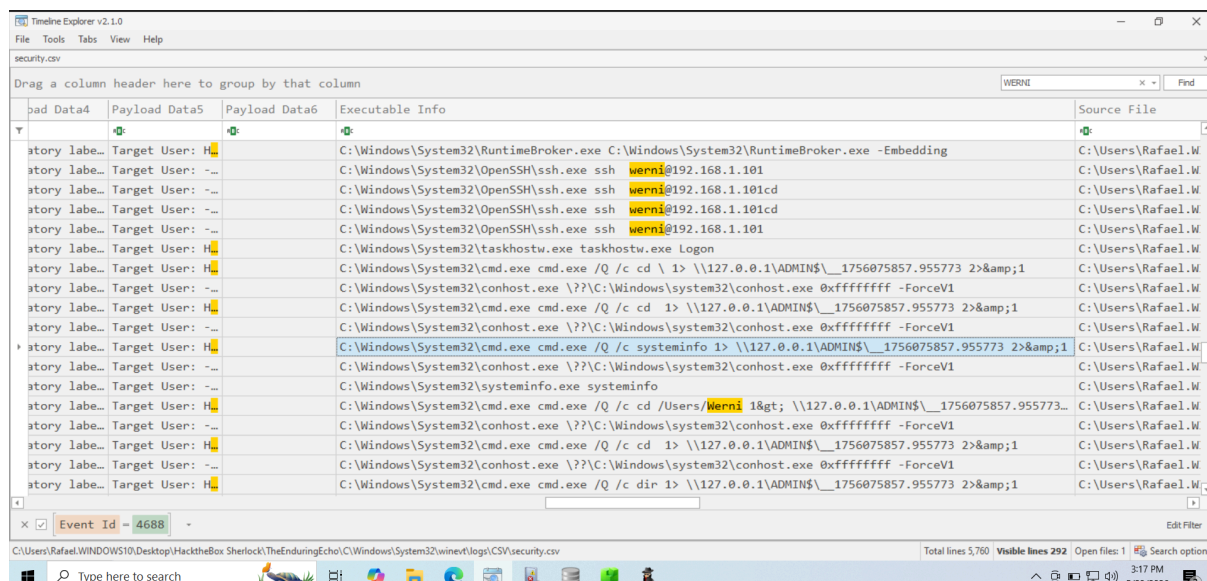
Command line: -f Security.evtx --csv CSV --csvf security.csv

Warning: Administrator privileges not found!

CSV output will be saved to CSV\security.csv

Maps loaded: 453

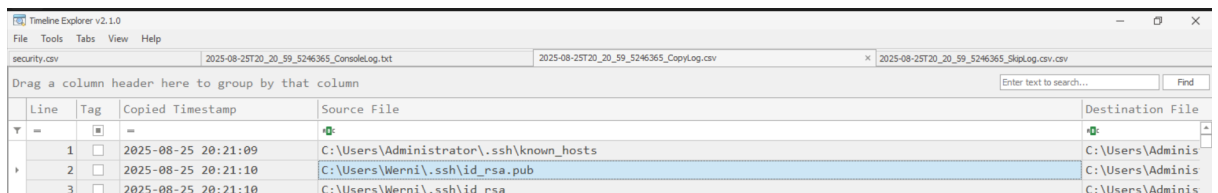
Processing C:\Users\Rafael.WINDOWS10\Desktop\HacktheBox Sherlock\TheEnduringEcho\C\Windows\System32\winevt\logs\Security.evtx...
Chunk count: 76, Iterating records...
Record # 68 (Event Record Id: 68): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessName"] not found!
Replacing with empty string
Record # 68 (Event Record Id: 68): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessId"] not found!
Replacing with empty string
Record # 69 (Event Record Id: 69): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessName"] not found!
Replacing with empty string
Record # 69 (Event Record Id: 69): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessId"] not found!
Replacing with empty string
Record # 70 (Event Record Id: 70): In map for event 4718, Property /Event/EventData/Data[@Name="ProcessName"] not found!
```



| Payload Data4 | Payload Data5     | Payload Data6 | Executable Info                                                                                       | Source File          |
|---------------|-------------------|---------------|-------------------------------------------------------------------------------------------------------|----------------------|
| atory labe... | Target User: H... |               | C:\Windows\System32\RuntimeBroker.exe C:\Windows\System32\RuntimeBroker.exe -Embedding                | C:\Users\Rafael.W... |
| atory labe... | Target User: ...  |               | C:\Windows\System32\OpenSSH\ssh.exe ssh werni@192.168.1.101                                           | C:\Users\Rafael.W... |
| atory labe... | Target User: ...  |               | C:\Windows\System32\OpenSSH\ssh.exe ssh werni@192.168.1.101cd                                         | C:\Users\Rafael.W... |
| atory labe... | Target User: ...  |               | C:\Windows\System32\OpenSSH\ssh.exe ssh werni@192.168.1.101cd                                         | C:\Users\Rafael.W... |
| atory labe... | Target User: ...  |               | C:\Windows\System32\OpenSSH\ssh.exe ssh werni@192.168.1.101                                           | C:\Users\Rafael.W... |
| atory labe... | Target User: H... |               | C:\Windows\System32\taskhostw.exe taskhostw.exe Logon                                                 | C:\Users\Rafael.W... |
| atory labe... | Target User: ...  |               | C:\Windows\System32\cmd.exe cmd.exe /Q /c cd \ 1> \\127.0.0.1\ADMIN\$_1756075857.955773 2>&1          | C:\Users\Rafael.W... |
| atory labe... | Target User: ...  |               | C:\Windows\System32\conhost.exe ???C:\Windows\system32\conhost.exe 0xffffffff -ForceV1                | C:\Users\Rafael.W... |
| atory labe... | Target User: H... |               | C:\Windows\System32\cmd.exe cmd.exe /Q /c cd 1> \\127.0.0.1\ADMIN\$_1756075857.955773 2>&1            | C:\Users\Rafael.W... |
| atory labe... | Target User: ...  |               | C:\Windows\System32\conhost.exe ???C:\Windows\system32\conhost.exe 0xffffffff -ForceV1                | C:\Users\Rafael.W... |
| atory labe... | Target User: ...  |               | C:\Windows\System32\conhost.exe ???C:\Windows\system32\conhost.exe 0xffffffff -ForceV1                | C:\Users\Rafael.W... |
| atory labe... | Target User: H... |               | C:\Windows\System32\cmd.exe cmd.exe /Q /c systeminfo 1> \\127.0.0.1\ADMIN\$_1756075857.955773 2>&1    | C:\Users\Rafael.W... |
| atory labe... | Target User: ...  |               | C:\Windows\System32\conhost.exe ???C:\Windows\system32\conhost.exe 0xffffffff -ForceV1                | C:\Users\Rafael.W... |
| atory labe... | Target User: ...  |               | C:\Windows\System32\systeminfo.exe systeminfo                                                         | C:\Users\Rafael.W... |
| atory labe... | Target User: H... |               | C:\Windows\System32\cmd.exe cmd.exe /Q /c cd /Users/werni 1> \\127.0.0.1\ADMIN\$_1756075857.955773... | C:\Users\Rafael.W... |
| atory labe... | Target User: ...  |               | C:\Windows\System32\conhost.exe ???C:\Windows\system32\conhost.exe 0xffffffff -ForceV1                | C:\Users\Rafael.W... |
| atory labe... | Target User: H... |               | C:\Windows\System32\cmd.exe cmd.exe /Q /c cd 1> \\127.0.0.1\ADMIN\$_1756075857.955773 2>&1            | C:\Users\Rafael.W... |
| atory labe... | Target User: ...  |               | C:\Windows\System32\conhost.exe ???C:\Windows\system32\conhost.exe 0xffffffff -ForceV1                | C:\Users\Rafael.W... |
| atory labe... | Target User: H... |               | C:\Windows\System32\cmd.exe cmd.exe /Q /c dir 1> \\127.0.0.1\ADMIN\$_1756075857.955773 2>&1           | C:\Users\Rafael.W... |

For this question I used EvtxECmd.exe to parse the Security.evtx Windows log file obtained from the C:\Windows\System32\winevt\logs directory and

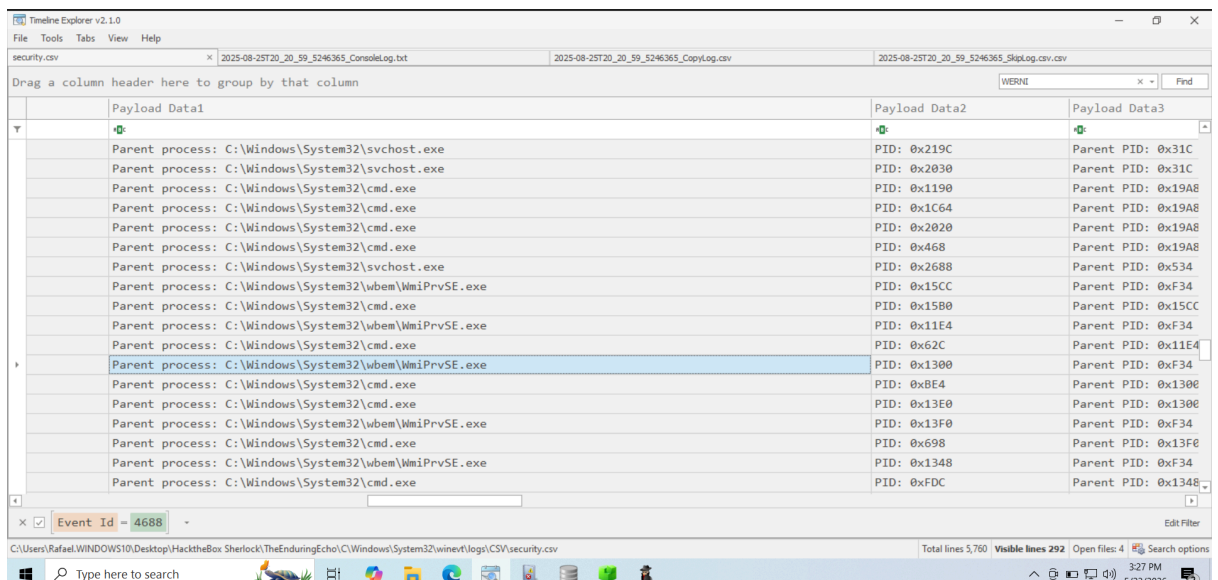
I filtered for process creation by filtering event ID to 4688 and then I filtered out for Werni which is the user seen from the other logs.



| Line | Tag | Copied Timestamp    | Source File                             | Destination File |
|------|-----|---------------------|-----------------------------------------|------------------|
| 1    |     | 2025-08-25 20:21:09 | C:\Users\Administrator\.ssh\known_hosts | C:\Users\Adminis |
| 2    |     | 2025-08-25 20:21:10 | C:\Users\Werni\.ssh\id_rsa.pub          | C:\Users\Adminis |
| 3    |     | 2025-08-25 20:21:10 | C:\Users\Werni\.ssh\id_rsa              | C:\Users\Adminis |

And then not long after the SSH command getting into werni at 192.168.1.101, there is a suspicious command after cd which is the first non cd command, which is **systeminfo** which is the answer to this question

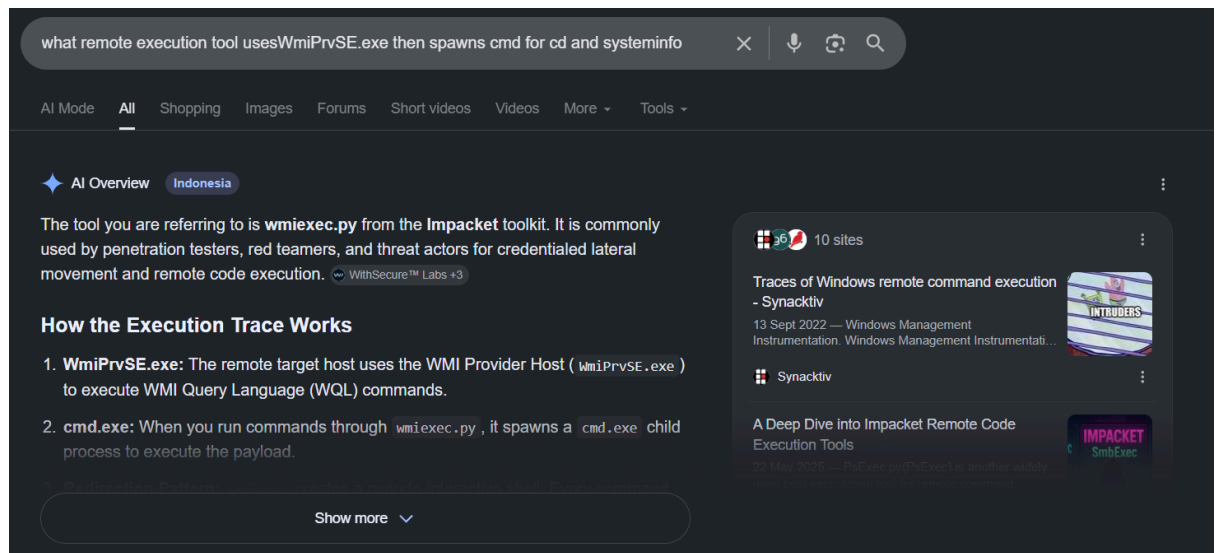
## 2. Which parent process (full path) spawned the attacker's commands?



| Payload Data1                                         | Payload Data2 | Payload Data3      |
|-------------------------------------------------------|---------------|--------------------|
| Parent process: C:\Windows\System32\svchost.exe       | PID: 0x219C   | Parent PID: 0x31C  |
| Parent process: C:\Windows\System32\svchost.exe       | PID: 0x2030   | Parent PID: 0x31C  |
| Parent process: C:\Windows\System32\cmd.exe           | PID: 0x1190   | Parent PID: 0x19A8 |
| Parent process: C:\Windows\System32\cmd.exe           | PID: 0x1C64   | Parent PID: 0x19A8 |
| Parent process: C:\Windows\System32\cmd.exe           | PID: 0x2020   | Parent PID: 0x19A8 |
| Parent process: C:\Windows\System32\cmd.exe           | PID: 0x468    | Parent PID: 0x19A8 |
| Parent process: C:\Windows\System32\svchost.exe       | PID: 0x2688   | Parent PID: 0x534  |
| Parent process: C:\Windows\System32\wbem\WmiPrvSE.exe | PID: 0x15CC   | Parent PID: 0xF34  |
| Parent process: C:\Windows\System32\cmd.exe           | PID: 0x15B0   | Parent PID: 0x15CC |
| Parent process: C:\Windows\System32\wbem\WmiPrvSE.exe | PID: 0x11E4   | Parent PID: 0xF34  |
| Parent process: C:\Windows\System32\cmd.exe           | PID: 0x62C    | Parent PID: 0x11E4 |
| Parent process: C:\Windows\System32\wbem\WmiPrvSE.exe | PID: 0x1300   | Parent PID: 0xF34  |
| Parent process: C:\Windows\System32\cmd.exe           | PID: 0xBE4    | Parent PID: 0x1300 |
| Parent process: C:\Windows\System32\cmd.exe           | PID: 0x13E0   | Parent PID: 0x1300 |
| Parent process: C:\Windows\System32\wbem\WmiPrvSE.exe | PID: 0x13F0   | Parent PID: 0xF34  |
| Parent process: C:\Windows\System32\cmd.exe           | PID: 0x698    | Parent PID: 0x13F0 |
| Parent process: C:\Windows\System32\wbem\WmiPrvSE.exe | PID: 0x1348   | Parent PID: 0xF34  |
| Parent process: C:\Windows\System32\cmd.exe           | PID: 0xFDC    | Parent PID: 0x1348 |

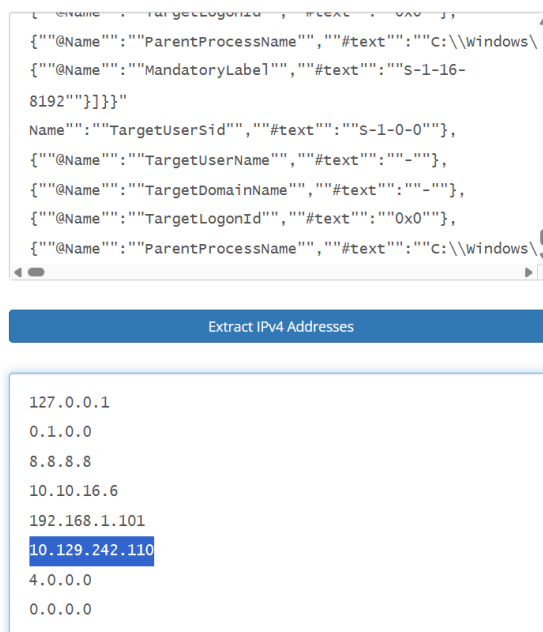
Still within the same row as the previous question, I just switched to the column of PayloadData1 which is the Parent Process Name and which allowed and spawned the attacker's command, and it is **C:\Windows\System32\wbem\WmiPrvSE.exe**

## 3. Which remote-execution tool was most likely used for the attack?

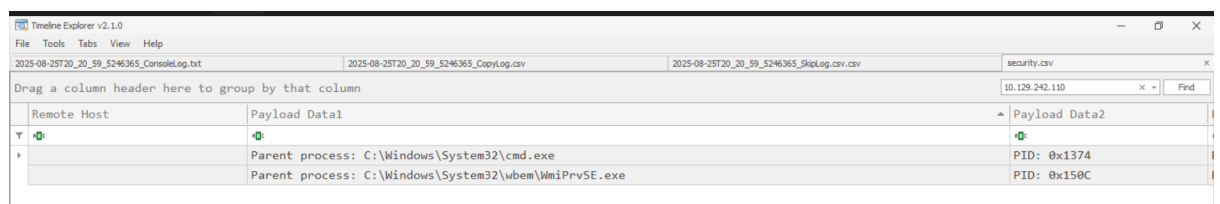


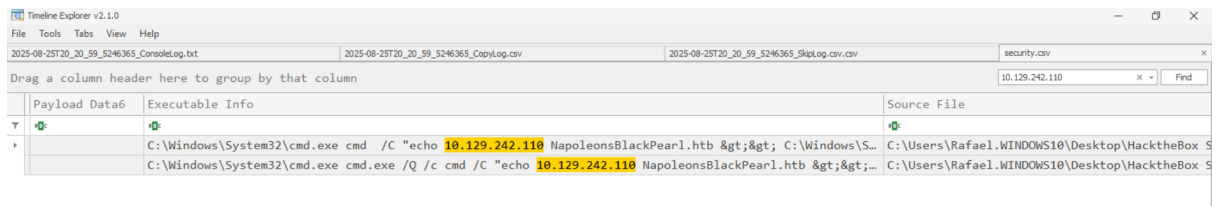
As Googled like the shown screenshot above, when using the WmiPrvSE.exe to spawn cmd for cd and systeminfo is [wmiexec.py](#)

#### 4. What was the attacker's IP address?



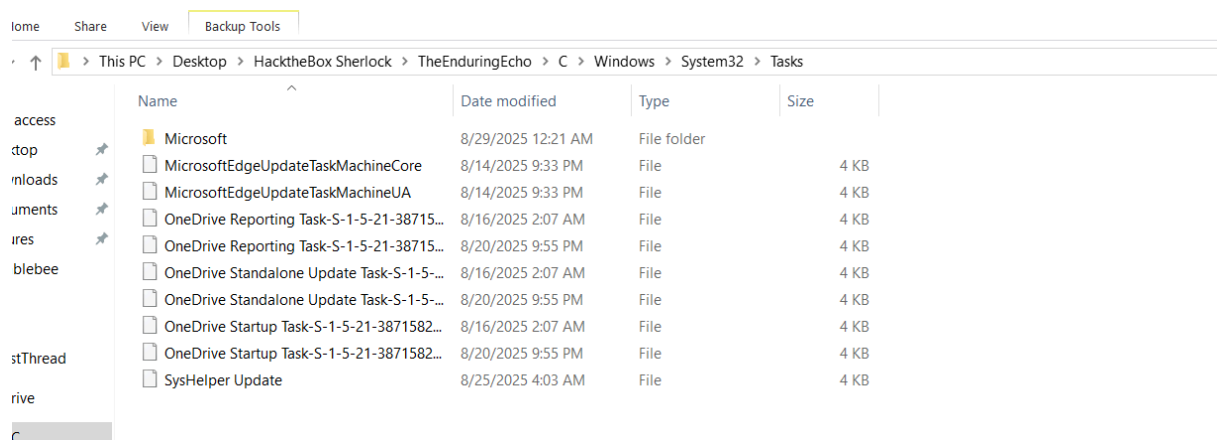
I searched up an IPv4 extractor program from the internet. In this case I am using [ipvoid.com](#). And then I found an oddly looking IP address, and then I searched it up on the security.csv



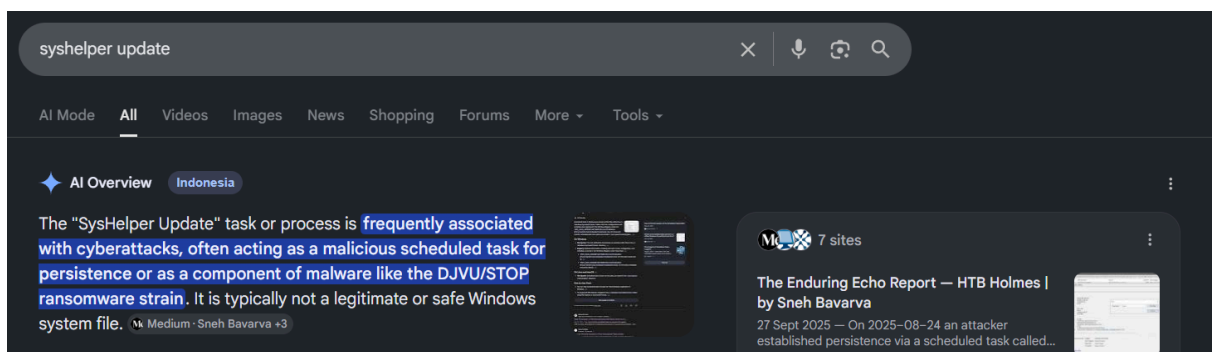


And yes, we can see that its parent's process are the cmd.exe and WmiPrvSE.exe So the answer is **10.129.242.110**

5. The attacker established multiple persistence mechanisms. What is set as the name of the earliest one created?



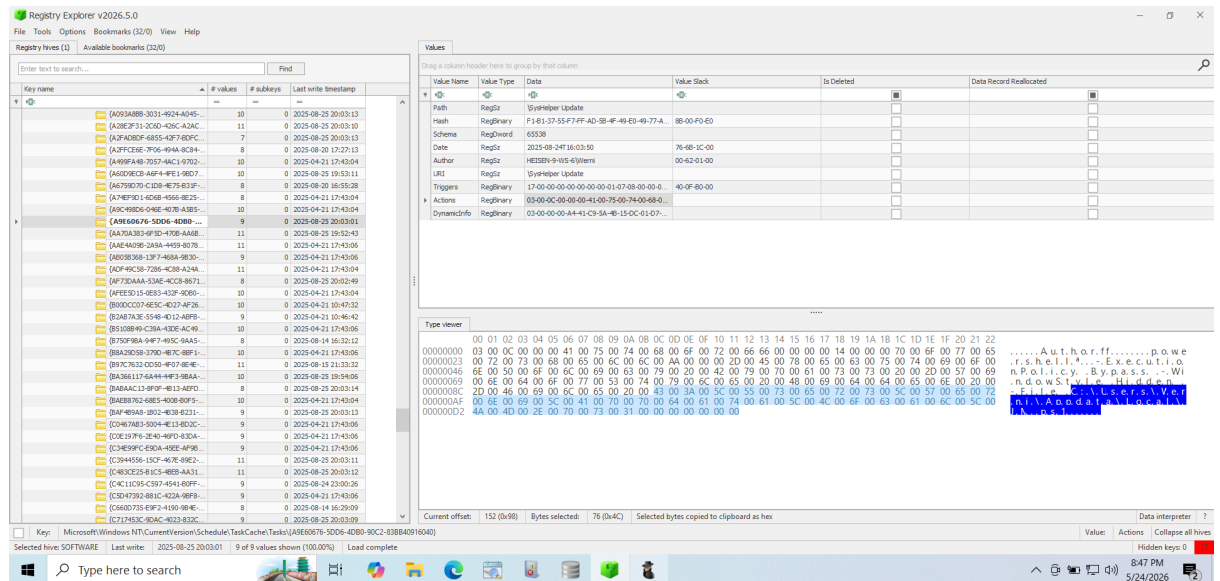
I went to the C:\Windows\System32\Tasks folder where the Scheduled Tasks. Then I saw a suspicious looking one which is SysHelper Update. I then Googled what is **SysHelper Update**.



And it is associated with scheduled tasks for persistence.

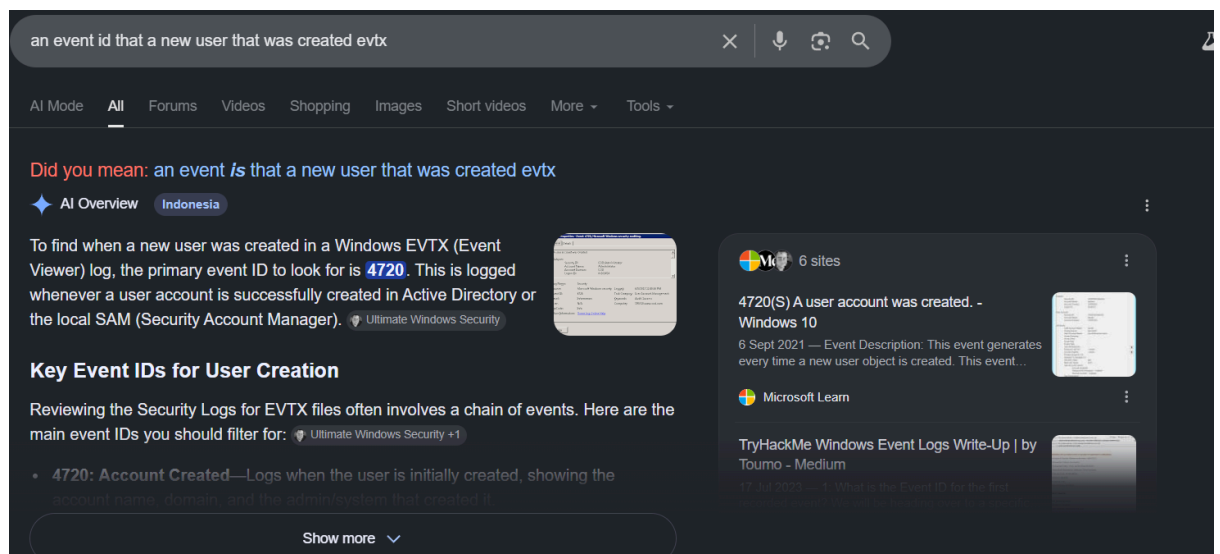
6. Identify the script executed by the persistence mechanism.

# Rafael Angelo Christianto

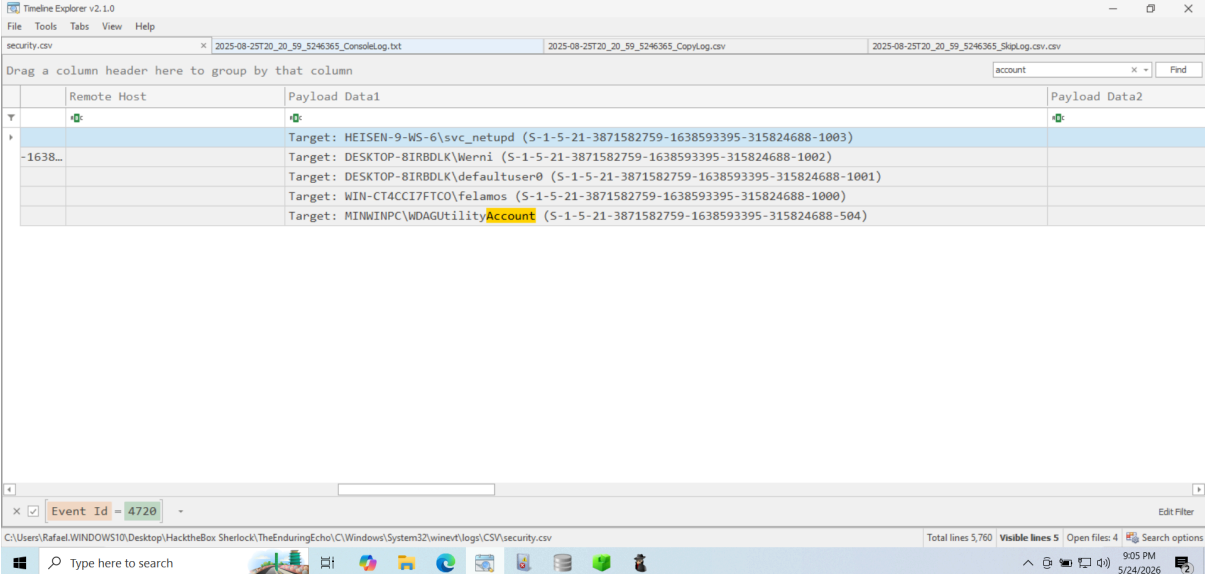


Here we can see the script executed in the Actions row still on the ID of the SysHelper Update since its the one who executed and highlighted in the screenshot above we can see the script executed until the drive letter so the answer is **C:\Users\Werni\AppData\Local\JM.ps1**

## 7. What local account did the attacker create?



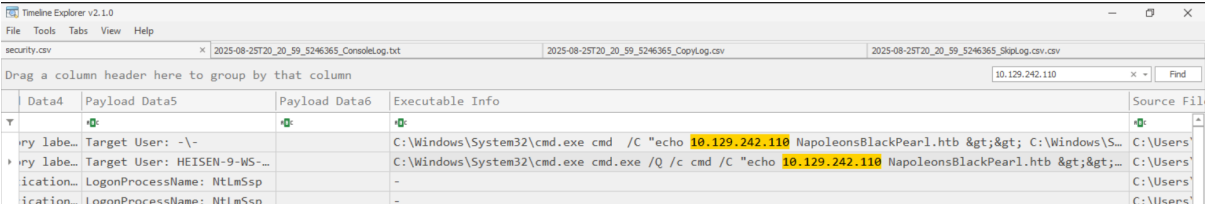
The Event ID in an evtx for a new user creation is 4720 so I will filter to that



| Remote Host                                                                          | Payload Data1 | Payload Data2 |
|--------------------------------------------------------------------------------------|---------------|---------------|
| Target: HEISEN-9-WS-6\svc_netupd (S-1-5-21-3871582759-1638593395-315824688-1003)     |               |               |
| Target: DESKTOP-8IRBDLK\Werni (S-1-5-21-3871582759-1638593395-315824688-1002)        |               |               |
| Target: DESKTOP-8IRBDLK\defaultuser0 (S-1-5-21-3871582759-1638593395-315824688-1001) |               |               |
| Target: WIN-CT4CCI7FTCO\felamos (S-1-5-21-3871582759-1638593395-315824688-1000)      |               |               |
| Target: MINWINPC\WDAGUtilityAccount (S-1-5-21-3871582759-1638593395-315824688-504)   |               |               |

Then I saw someone created called **svc\_netupd** under HEISEN-9-WS-6 which is the answer to this question

## 8. What domain name did the attacker use for credential exfiltration?

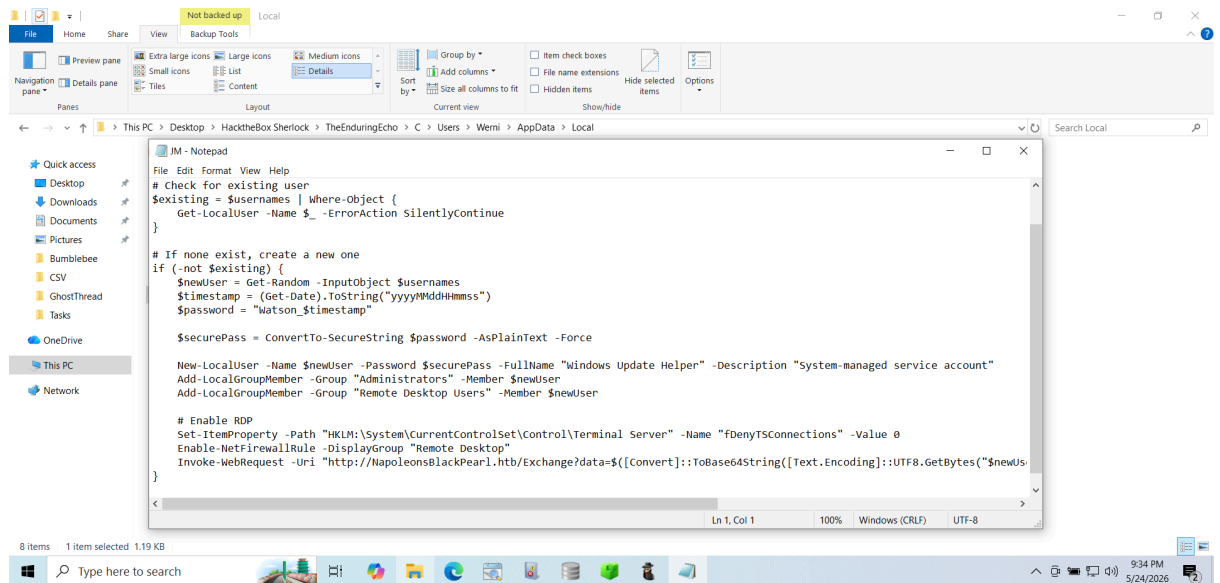


| Data4                        | Payload Data5 | Payload Data6 | Executable Info                                                                                                        | Source File  |
|------------------------------|---------------|---------------|------------------------------------------------------------------------------------------------------------------------|--------------|
| Target User: -\-             |               |               | C:\Windows\System32\cmd.exe cmd /c "echo 10.129.242.110 NapoleonsBlackPearl.htb &gt;&gt; C:\Windows\S...               | C:\Users\... |
| Target User: HEISEN-9-WS-... |               |               | C:\Windows\System32\cmd.exe cmd.exe /Q /c cmd /c "echo 10.129.242.110 NapoleonsBlackPearl.htb &gt;&gt; C:\Windows\S... | C:\Users\... |
| LogonProcessName: NtLmSsp    |               |               | -                                                                                                                      | C:\Users\... |
| LogonProcessName: NtLmSsp    |               |               | -                                                                                                                      | C:\Users\... |

I searched in the security.csv the attacker's IP address since it would correlate to the domain name the attacker use to exfiltrate the credentials and there the answer is **NapoleonsBlackPearl.htb**

## 9. What password did the attacker's script generate for the newly created user?

## Rafael Angelo Christianto



For this question I tried to look for the JM.ps1 file from the previous question since under the Users folder there is also the Werni Folder and under C:\Users\Werni\AppData\Local, the password template is “Watson\_\$timestamp” and the timestamp is when the user was created

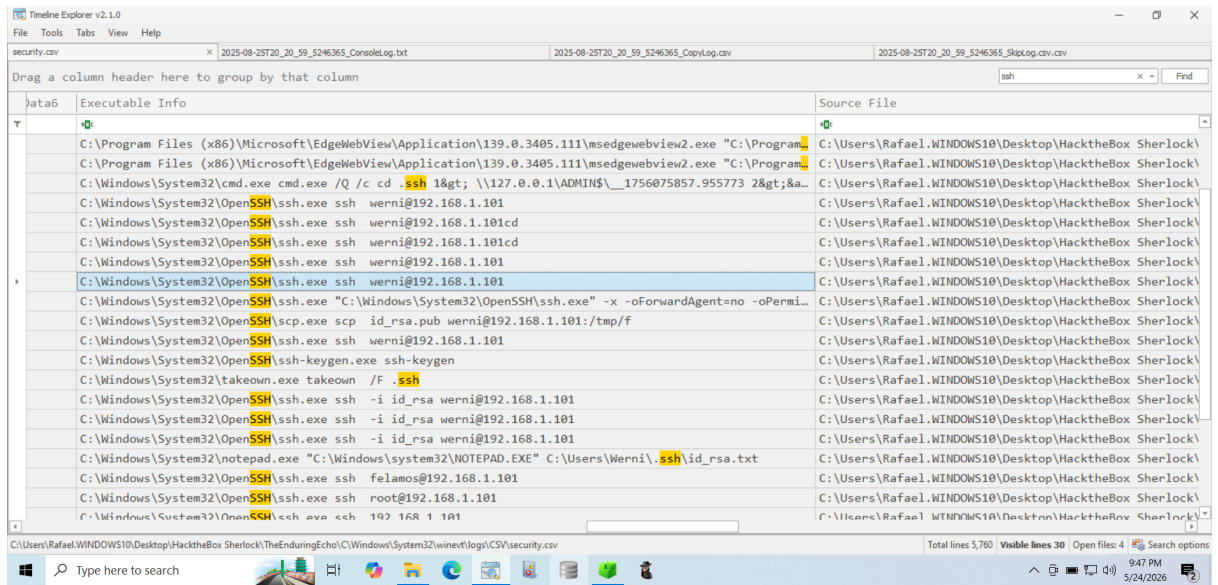
| Line | Tag | Record Number | Event Record Id | Time Created        | Event Id | Level     | Provider                            | Channel  | Process |
|------|-----|---------------|-----------------|---------------------|----------|-----------|-------------------------------------|----------|---------|
| 5093 |     | 5093          | 5093            | 2025-08-25 19:52:38 | 4798     | LogAlways | Microsoft-Windows-Security-Auditing | Security |         |
| 5031 |     | 5031          | 5031            | 2025-08-25 19:51:38 | 4798     | LogAlways | Microsoft-Windows-Security-Auditing | Security |         |
| 4942 |     | 4942          | 4942            | 2025-08-25 19:50:26 | 4798     | LogAlways | Microsoft-Windows-Security-Auditing | Security |         |
| 4652 |     | 4652          | 4652            | 2025-08-24 23:13:59 | 4798     | LogAlways | Microsoft-Windows-Security-Auditing | Security |         |
| 4464 |     | 4464          | 4464            | 2025-08-24 23:05:09 | 4738     | LogAlways | Microsoft-Windows-Security-Auditing | Security |         |
| 4463 |     | 4463          | 4463            | 2025-08-24 23:05:09 | 4738     | LogAlways | Microsoft-Windows-Security-Auditing | Security |         |
| 4465 |     | 4465          | 4465            | 2025-08-24 23:05:09 | 4724     | LogAlways | Microsoft-Windows-Security-Auditing | Security |         |
| 4462 |     | 4462          | 4462            | 2025-08-24 23:05:09 | 4722     | LogAlways | Microsoft-Windows-Security-Auditing | Security |         |
| 4461 |     | 4461          | 4461            | 2025-08-24 23:05:09 | 4720     | LogAlways | Microsoft-Windows-Security-Auditing | Security |         |

Highlighted above is the row when the account was created and I have to subtract it by 7 to turn it to UTC so the format becomes 20250824160509 combining it with the format we get **Watson\_20250824160509**

10.What was the IP address of the internal system the attacker pivoted to?

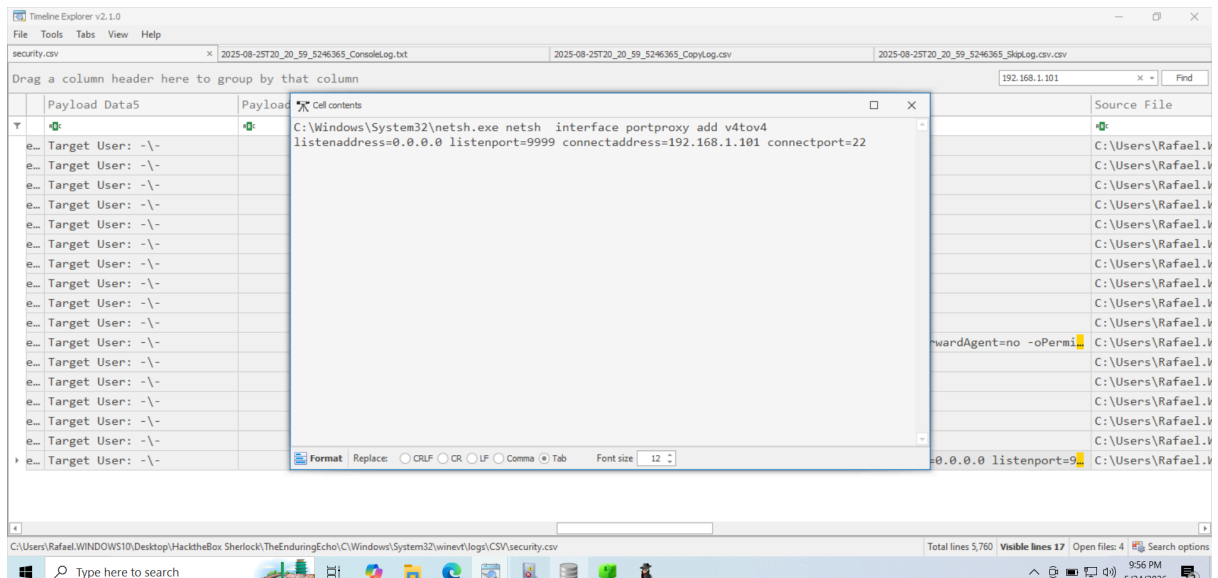


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Here we can see the IP 192.168.1.101 and seeing from the IP it looks like an internal pivot to the user werni. Therefore, we can the IP is **192.168.1.101**

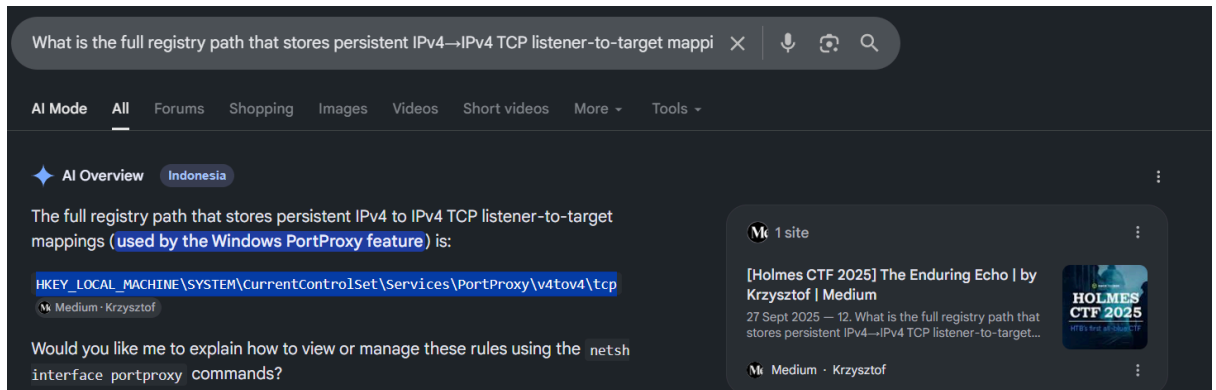
11.Which TCP port on the victim was forwarded to enable the pivot?



Here I searched the IP 192.168.1.101 to filter out again more narrow and the answer is not 22 since that is the connect port this question is at the victim's side where it is listening which is at port **9999**



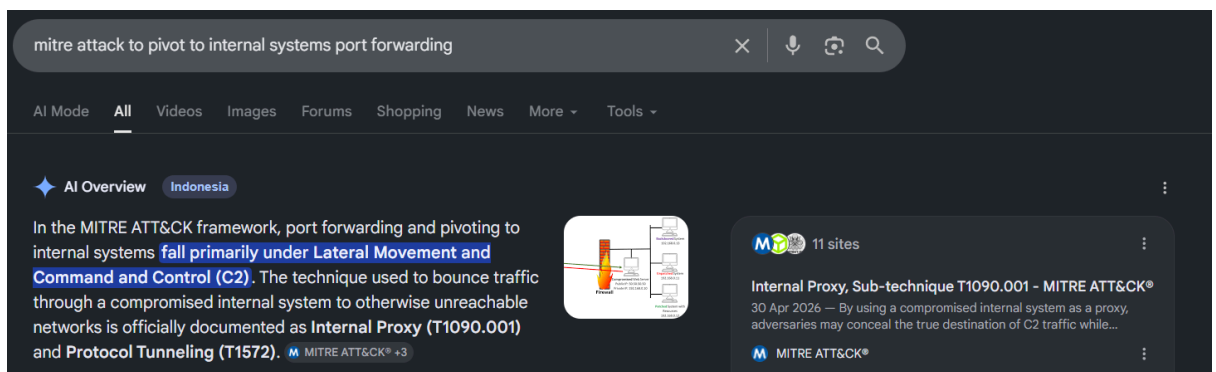
12. What is the full registry path that stores persistent IPv4→IPv4 TCP listener-to-target mappings?



Here I will Google the full registry path that stores persistent TCP IPv4 connections listener-to-target mappings which is

**HKLM\SYSTEM\CurrentControlSet\Services\PortProxy\v4tov4\tcp**

13. What is the MITRE ATT&CK ID associated with the previous technique used by the attacker to pivot to the internal system?



I Googled what would be the suitable MITRE ATT&CK ID for this Sherlock's case and since it also did pivoting with SSH and port forwarding it can be considered as Internal Proxy (**T1090.001**)

Home > Techniques > Enterprise > Proxy > Internal Proxy

## Proxy: Internal Proxy

Other sub-techniques of Proxy (4)

Adversaries may use an internal proxy to direct command and control traffic between two or more systems in a compromised environment. Many tools exist that enable traffic redirection through proxies or port redirection, including [HTRAN](#), [ZXProxy](#), and [ZXPortMap](#).<sup>[1]</sup> Adversaries use internal proxies to manage command and control communications inside a compromised environment, to reduce the number of simultaneous outbound network connections, to provide resiliency in the face of connection loss, or to ride over existing trusted communications paths between infected systems to avoid suspicion. Internal proxy connections may use common peer-to-peer (p2p) networking protocols, such as SMB, to better blend in with the environment.

By using a compromised internal system as a proxy, adversaries may conceal the true destination of C2 traffic while reducing the need for numerous connections to external systems.

ID: T1090.001

Sub-technique of: [T1090](#)

① **Tactic:** [Command and Control](#)

① **Platforms:** ESXi, Linux, Network Devices, Windows, macOS

Version: 1.2

Created: 14 March 2020

Last Modified: 12 May 2026

[Version Permalink](#)

14. Before the attack, the administrator configured Windows to capture command line details in the event logs. What command did they run to achieve this?

The screenshot shows a search result for the query "administrator configured Windows to capture command line details in the event logs, how it is done". The results page includes an "AI Overview" section with a summary: "To log command-line details in Windows Event Logs, administrators must enable process tracking (Event ID 4688) and explicitly configure the system to include command-line arguments. You can do this via command prompt by configuring the local security policy and registry values." Below this is a "Step-by-Step Command Line Configuration" section. Step 1, "Enable Audit Process Creation:", shows a command prompt window with the command: `cmd`  
`ditpol /set /subcategory:"Process Creation" /success:enable /failure:enable`. Step 2, "Enable Command Line Inclusion in Events:", shows a command prompt window with the command: `reg add "HKLM\Software\Microsoft\Windows\CurrentVersion\Policies\System\Audit" /v ProcessCreationIncludeCmdLine_Enabled /t REG_DWORD /d 1 /f`. The right sidebar shows a list of 6 sites, including "Audit Process Creation And Command Line - Elastic", "Troubleshooting Missing Process Command Lines in Windows ...", "Solved: Re: CMD Command Line Logging - Splunk Community", and "Building a Detection Foundation: Part 2 - Windows...".

I Googled on how to achieve this and they have to add something to the registry and the answer is **reg add**  
**"HKLM\Software\Microsoft\Windows\CurrentVersion\Policies\System\Audit" /v ProcessCreationIncludeCmdLine\_Enabled /t REG\_DWORD /d 1 /f**



**You have solved Holmes 2025 3: The Enduring Echo!**

Congratulations  **rafaelangelo** best of luck in capturing flags ahead!

|               |                    |                |
|---------------|--------------------|----------------|
| <b>#323</b>   | <b>24 May 2026</b> | <b>Retired</b> |
| Sherlock Rank | Solve Date         | Sherlock State |